Statistical Inference: Estimation

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Estimation vs Hypothesis Testing

Estimation:

- Involves estimating the value of a population parameter based on sample data.
- Example: Estimating the population mean μ using a sample mean $\bar{X}$.

Hypothesis Testing:

- Involves testing a claim about a population parameter using sample data.
- Example: Testing if the population mean $\mu = 50$ based on sample data.

Estimates and Estimators

Estimate:

- A specific value derived from the sample data that serves as the approximation of the population parameter.
- Example: The sample mean $\bar{X}$ is an estimate of the population mean μ.

Estimator:

- A rule or formula used to compute an estimate from sample data.
- Example: The formula $\bar{X}$ is an estimator of μ.

Symbols for Population Parameters and Sample Parameters

Population Parameters:

- μ – Population mean
- σ^2 – Population variance
- σ – Population standard deviation
- N – Population size

Sample Parameters:

- $\bar{X}$ – Sample mean
- s^2 – Sample variance
- s – Sample standard deviation
- n – Sample size

Point Estimate vs Interval Estimate

Point Estimate:

- A single value used to estimate the population parameter.
- Example: The sample mean $\bar{X} = 45$ is a point estimate of the population mean μ.

Interval Estimate:

- A range of values used to estimate the population parameter.
- Example: A 95% confidence interval $[40, 50]$ for the population mean μ.

Criteria for a Good Estimate

A good estimator should possess the following four properties:

- **Unbiasedness:** The estimator is unbiased if its expected value equals the true population parameter.

$$E(\hat{\theta}) = \theta$$

- **Consistency:** The estimator is consistent if, as the sample size n increases, the estimator tends to the true population parameter.

$$\hat{\theta}_n \xrightarrow{p} \theta \quad \text{as} \quad n \to \infty$$

- **Efficiency:** An efficient estimator has the smallest possible variance among all unbiased estimators.
- **Sufficiency:** An estimator is sufficient if it uses all the information in the sample that is relevant to estimating the parameter.

Proof of Unbiasedness of Sample Mean and Sample Variance

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Unbiasedness of Sample Mean

Let $X_1, X_2, \ldots, X_n$ be a random sample drawn from a population with mean μ and variance σ^2. The sample mean is defined as:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

We need to show that $\mathbb{E}[\bar{X}] = \mu$.
By the linearity of expectation:

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[X_i]$$

Since $X_1, X_2, \ldots, X_n$ are i.i.d. with $\mathbb{E}[X_i] = \mu$, we get:

$$\mathbb{E}[\bar{X}] = \frac{1}{n} \cdot n\mu = \mu$$

Hence, the sample mean is an unbiased estimator of μ.

Unbiasedness of Sample Variance

Let $X_1, X_2, \ldots, X_n$ be a random sample drawn from a population with mean μ and variance σ^2. The sample variance is defined as:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2$$

We need to show that $\mathbb{E}[s^2] = \sigma^2$.
First, expand the squared deviations:

$$\sum_{i=1}^{n} (X_i - \bar{X})^2 = \sum_{i=1}^{n} (X_i - \mu)^2 - n(\bar{X} - \mu)^2$$

Therefore, the sample variance becomes:

$$s^2 = \frac{1}{n-1} \left(\sum_{i=1}^{n} (X_i - \mu)^2 - n(\bar{X} - \mu)^2 \right)$$

Expectation of Sample Variance

Now, take the expectation of s^2:

$$\mathbb{E}[s^2] = \frac{1}{n-1}\left(\mathbb{E}\left[\sum_{i=1}^{n}(X_i-\mu)^2\right] - n\mathbb{E}\left[(\bar{X}-\mu)^2\right]\right)$$

- The first term, $\mathbb{E}\left[\sum_{i=1}^{n}(X_i-\mu)^2\right]$, is simply $n\sigma^2$, because $\mathbb{E}[(X_i-\mu)^2] = \sigma^2$ for each i.

$$\mathbb{E}\left[\sum_{i=1}^{n}(X_i-\mu)^2\right] = n\sigma^2$$

- The second term, $\mathbb{E}\left[(\bar{X}-\mu)^2\right]$, is the variance of the sample mean, which is $\frac{\sigma^2}{n}$:

$$\mathbb{E}\left[(\bar{X}-\mu)^2\right] = \frac{\sigma^2}{n}$$

Final Simplification

Substituting these into the equation for $\mathbb{E}[s^2]$:

$$\mathbb{E}[s^2] = \frac{1}{n-1}\left(n\sigma^2 - n \cdot \frac{\sigma^2}{n}\right)$$

Simplifying:

$$\mathbb{E}[s^2] = \frac{1}{n-1} \cdot (n-1)\sigma^2 = \sigma^2$$

Therefore, the sample variance is an unbiased estimator of σ^2.

Consistency and Proof for Sample Mean $\bar{X}$

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Definition of Consistency

Consistency

An estimator $\hat{\theta}_n$ is said to be **consistent** for a parameter θ if:

$$\hat{\theta}_n \xrightarrow{P} \theta \quad \text{as} \quad n \to \infty$$

which means, for any $\varepsilon > 0$,

$$\lim_{n\to\infty} P(|\hat{\theta}_n - \theta| > \varepsilon) = 0.$$

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- If $\mathbb{E}[\hat{\theta}_n] \to \theta$ and $\text{Var}(\hat{\theta}_n) \to 0$ as $n \to \infty$, then $\hat{\theta}_n$ is consistent.

Proof that $\bar{X}$ is a Consistent Estimator of μ

Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables with:

- $\mathbb{E}[X_i] = \mu$
- $\mathrm{Var}(X_i) = \sigma^2$

Define the sample mean:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

Step 1: Expectation of $\bar{X}$

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \frac{n\mu}{n} = \mu$$

- Thus, $\bar{X}$ is an **unbiased estimator** of μ.

Step 2: Variance of $\bar{X}$

$$\mathrm{Var}(\bar{X}) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

- As $n \to \infty$, $\mathrm{Var}(\bar{X}) \to 0$.

Conclusion

Since:

- $\mathbb{E}[\bar{X}] = \mu$,
- $\text{Var}(\bar{X}) \to 0$ as $n \to \infty$,

therefore, by definition, $\bar{X}$ is a **consistent estimator** of μ.

$$\boxed{\bar{X} \text{ is consistent for } \mu}$$

Efficiency of Estimators

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Definition of Efficiency

Efficiency

An estimator $\hat{\theta}_1$ is said to be **more efficient** than another estimator $\hat{\theta}_2$ if:

$$\mathrm{Var}(\hat{\theta}_1) < \mathrm{Var}(\hat{\theta}_2)$$

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- **Relative Efficiency**:

$$\text{Relative Efficiency} = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)}$$

- Higher relative efficiency implies better performance.

Example Problem

- Estimator 1: $\hat{\theta}_1 = \bar{X}$
- Estimator 2: $\hat{\theta}_2 = \dfrac{X_1 + 2X_2}{3}$
- Assume $X_1, X_2, \ldots, X_n$ are i.i.d with mean μ and variance σ^2.

Solution - Variances

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

$$\text{Var}\left(\frac{X_1 + 2X_2}{3}\right) = \frac{1}{9}(\sigma^2 + 4\sigma^2) = \frac{5\sigma^2}{9}$$

Solution - Efficiency Comparison

Since:

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} \quad \text{and} \quad \text{Var}(\hat{\theta}_2) = \frac{5\sigma^2}{9}$$

- $\bar{X}$ has smaller variance when $n > \frac{9}{5}$.
- Thus, $\bar{X}$ is more efficient.

Relative Efficiency Calculation

$$\text{Relative Efficiency} = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\bar{X})} = \frac{5n}{9}$$

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Example: For $n = 3$,

$$\text{Relative Efficiency} = \frac{5 \times 3}{9} = \frac{5}{3} \approx 1.666$$

$$\boxed{\bar{X} \text{ is 1.666 times more efficient than } \hat{\theta}_2}$$

Sufficiency of a Statistic

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Definition of Sufficiency

Sufficiency

A statistic $T(X_1, X_2, \ldots, X_n)$ is called **sufficient** for a parameter θ if the conditional distribution of $(X_1, X_2, \ldots, X_n)$ given $T(X)$ does not depend on θ.

Definition of Sufficiency

Sufficiency

A statistic $T(X_1, X_2, \ldots, X_n)$ is called **sufficient** for a parameter θ if the conditional distribution of $(X_1, X_2, \ldots, X_n)$ given $T(X)$ does not depend on θ.

- Intuitively: $T(X)$ summarizes all the information about θ.

Example Problem

Given:

$$f(x;\theta) = \theta x^{\theta-1}, \quad 0 < x < 1, \quad \theta > 0$$

Show that $T(X) = \sum_{i=1}^{n} \ln X_i$ is a sufficient statistic for θ.

Solution - Joint PDF

Since $X_1, X_2, \ldots, X_n$ are i.i.d:

$$f(x_1, \ldots, x_n; \theta) = \prod_{i=1}^{n} \theta x_i^{\theta-1} = \theta^n \left(\prod_{i=1}^{n} x_i \right)^{\theta-1}$$

$$= \theta^n \exp\left((\theta - 1) \sum_{i=1}^{n} \ln x_i \right)$$

Solution - Factorization

- The joint pdf depends on the sample only through $\sum \ln X_i$.
- Thus, $T(X) = \sum \ln X_i$ is sufficient for θ.

$$\boxed{\text{Thus, } T(X) \text{ is sufficient for } \theta}$$

Example: Normal Distribution

Suppose:

$$X_1, X_2, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$$

where σ^2 is known.

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Suppose:

$$X_1, X_2, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$$

where σ^2 is known.
We want to show that $\bar{X}$ is a **sufficient statistic** for μ.

Solution - Joint PDF

The joint pdf is:

$$f(x_1, \ldots, x_n; \mu) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

$$= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2\right)$$

Simplifying the Exponent

Expand:

$$\sum(x_i - \mu)^2 = \sum x_i^2 - 2\mu \sum x_i + n\mu^2$$

Thus:

$$f(x_1, \ldots, x_n; \mu) = g(x_1, \ldots, x_n) \times h\left(\sum x_i, \mu\right)$$

Simplifying the Exponent

Expand:

$$\sum (x_i - \mu)^2 = \sum x_i^2 - 2\mu \sum x_i + n\mu^2$$

Thus:

$$f(x_1, \ldots, x_n; \mu) = g(x_1, \ldots, x_n) \times h\left(\sum x_i, \mu\right)$$

By the **Factorization Theorem**, $\sum x_i$ (or $\bar{X}$) is sufficient for μ.

Conclusion

$$\boxed{\bar{X} \text{ is sufficient for } \mu}$$

when σ^2 is known.

Bias-Variance Tradeoff and Mean Squared Error

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Mean Squared Error (MSE)

The **Mean Squared Error (MSE)** of an estimator $\hat{\theta}$ is a measure of the estimator's quality, combining both the variance and the bias.

$$\begin{aligned} \text{MSE}(\hat{\theta}) &= \mathbb{E}[(\hat{\theta} - \theta)^2] \\ &= \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}] + \mathbb{E}[\hat{\theta}] - \theta)^2] \\ &= \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2] + 2\mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])(\mathbb{E}[\hat{\theta}] - \theta)] + (\mathbb{E}[\hat{\theta}] - \theta)^2 \end{aligned}$$

The first term is the variance of the estimator:

$$\mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2] = \text{Var}(\hat{\theta})$$

The second term is:

$$2\mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])(\mathbb{E}[\hat{\theta}] - \theta)]$$

Since $\mathbb{E}[\hat{\theta}] - \theta$ is a constant, this term simplifies to zero:

$$\mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])(\mathbb{E}[\hat{\theta}] - \theta)] = 0$$

Thus, the second term vanishes.

Mean Squared Error (MSE) (Continued)

The third term is the square of the bias of the estimator:

$$(\mathbb{E}[\hat{\theta}] - \theta)^2 = (\text{Bias}(\hat{\theta}))^2$$

This term represents the squared difference between the expected value of the estimator and the true parameter. The Mean Squared Error (MSE) can be written as the sum of two components:

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \left(\text{Bias}(\hat{\theta})\right)^2$$

Key Points

A good estimator should have both low bias and low variance.

If the estimator is unbiased, $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta})$.

Bias-Variance Tradeoff

- **Bias:** Error from incorrect model assumptions (high bias = underfitting, low accuracy)
- **Variance:** Error from model sensitivity to data (high variance = overfitting, low precision)
- **Accuracy** improves as **bias** decreases
- **Precision** improves as **variance** decreases

Model Behavior

- **Underfitting:** High bias, low variance $\rightarrow$ poor accuracy, high generalization error
- **Overfitting:** Low bias, high variance $\rightarrow$ poor precision, unstable predictions
- **Good Model:** Balanced bias and variance $\rightarrow$ high accuracy and precision

Question: Simulation Study

Simulation Results:

Estimator	Avg Estimate	True Mean	Variance
A	10.5	10.0	1.2
B	9.8	10.0	0.5

Task:

(a) Calculate Bias and MSE.

(b) Which estimator is preferable?

Solution:

$$\text{Bias}_A = 10.5 - 10 = 0.5 \Rightarrow \text{MSE}_A = 0.5^2 + 1.2 = 1.45$$

$$\text{Bias}_B = 9.8 - 10 = -0.2 \Rightarrow \text{MSE}_B = 0.2^2 + 0.5 = 0.54$$

Conclusion:

- Estimator B is more accurate (lower MSE).
- Estimator B is also more precise.